

The Emergence of the Digital Euro: DLT-Inspired Design Principles and Tiered Architecture for Balancing Privacy and Regulatory Compliance

Ludovico Papalia
CIRSFID - Università di Bologna
Bologna, Italy
ludovico.papalia2@unibo.it

Abstract—The European Central Bank’s digital euro project confronts a structural tension between preserving individual privacy rights and ensuring regulatory compliance with anti-money laundering obligations. This paper examines the current state of the digital euro’s institutional development, technical architecture and privacy design, situating the European initiative within the broader international landscape of central bank digital currency implementations. Drawing upon official ECB documentation, a review of operational Central Bank Digital Currency deployments in selected jurisdictions, and empirical survey research on European privacy preferences, the analysis identifies a series of open problems that the current uniform pseudonymisation model leaves unresolved. A central observation of this analysis is that the ECB’s current design conditions privacy protection on the payment medium rather than on transaction risk, reserving strong anonymity guarantees for offline transactions while leaving online payments, where the overwhelming majority of economically significant activity occurs, subject only to structural pseudonymisation. Building on this analysis, the paper introduces the concept of graduated functional anonymity as a possible alternative framework warranting further investigation, outlining a three-tier privacy architecture that relocates the discriminating variable from the payment medium to the transaction amount, extending meaningful privacy protections to the online channel proportionally to transaction value and money laundering risk. The analysis further situates the proposed framework within the broader context of European digital sovereignty, arguing that a privacy architecture grounded in constitutional proportionality simultaneously serves as a contribution to European defensive resilience, limiting the strategic value of payment data in scenarios of financial infrastructure weaponisation. The proposal remains at a theoretical level, intended as a contribution to the ongoing scholarly and institutional debate rather than a prescriptive design recommendation.

Index Terms—Digital Euro, Central Bank Digital Currency, Distributed Ledger Technology, Blockchain, Privacy Preserving Technologies, Functional Anonymity, Zero Knowledge Proofs, Anti Money Laundering, Tiered Architecture, Digital Sovereignty, Threshold Cryptography, Post-Quantum Cryptography, GDPR, Financial Inclusion, Pseudonymisation, Proportionality Principle, Economic Warfare, Strategic Autonomy, Data Protection

I. INTRODUCTION

This paper examines the European Central Bank’s digital euro initiative through the lens of the unresolved tension

between transactional privacy and anti-money laundering compliance, a tension that, as the analysis will suggest, may prove determinative for the project’s long-term viability. The first part of the discussion situates the digital euro within its broader institutional, legal and geopolitical context, tracing the evolution of the ECB’s preparation phase, the architectural choices that distinguish the digital euro from blockchain-based payment infrastructures, and the comparative landscape of central bank digital currency implementations across jurisdictions ranging from China’s surveillance-oriented digital yuan to the more embryonic deployments in the Bahamas and Nigeria. The second part turns to the privacy-compliance dilemma itself, interrogating the limitations of the ECB’s current uniform pseudonymisation model in light of empirical evidence on European privacy preferences and the disappointing adoption trajectories observed in operational Central Bank Digital Currencies elsewhere. The third and final part introduces, in deliberately exploratory terms, analyses the concept of graduated functional anonymity as a possible alternative architectural framework, outlining a three-tier privacy model calibrated to transaction value and money laundering risk while acknowledging the substantial technical, regulatory and political challenges that any such approach would need to address. The central research question driving this analysis is whether the uniform pseudonymisation model currently adopted by the ECB can adequately reconcile European citizens’ documented privacy expectations with anti-money laundering obligations, or whether a graduated privacy architecture calibrated to transaction risk would better serve both imperatives. The methodology adopted in this article combines doctrinal legal analysis with systematic review of primary technical documentation. The source hierarchy proceeds from official ECB institutional documents, including the preparation phase closing report and the draft Regulation on the establishment of the digital euro, to secondary academic literature spanning legal informatics, cryptographic protocol design and monetary policy, to empirical survey data used to ground the analysis in observed citizen preferences. The scope of this study is explicitly theoretical and exploratory: the doc-

ument does not aim to offer a prescriptive technical blueprint, but rather to identify a little-explored conceptual space that merits further investigation. Since the very first analysis of the concept of a digital euro, all European institutions have based their analysis on the *concept first*, that is, building the technical architecture on the legal and social foundation, and not the other way around. This paper follows exactly the same principle, but also proposes a framework aimed at preserving privacy while maintaining full digital sovereignty. The primary sources consist of official ECB documentation, including the preparation phase closing report of October 2025, the draft Regulation on the establishment of the digital euro, and related technical annexes and stakeholder consultation materials. These institutional sources are supplemented by secondary literature spanning legal informatics, cryptographic protocol design and monetary policy. The paper also draws upon empirical survey data, in particular the Deutsche Bundesbank survey of 2024 and the ECB-commissioned user research of 2025, to ground its analysis in observed citizen preferences rather than purely theoretical assumptions. Selected international Central Bank Digital Currency (CBDC) implementations are referenced not as objects of comparative legal analysis but as empirical reference points illustrating the range of privacy design choices and their observable adoption consequences. On this basis, the paper proceeds in three parts: the first situates the digital euro within its institutional and technical context; the second identifies the privacy-compliance tension as the central open problem; and the third explores graduated functional anonymity as a possible architectural response warranting further investigation.

II. THE DIGITAL EURO: INSTITUTIONAL AND REGULATORY CONTEXT

The digital euro project, in fact, emerges from a confluence of strategic pressures that are actively reshaping the global monetary architecture. Competition among the United States, China and Europe has intensified around digital currency supremacy, with China's digital yuan operational since 2020 and over 260 million wallets activated as of 2023 [1]. The European Union faces what might be described as a dual threat: the potential displacement of the euro in digital transactions by foreign sovereign digital currencies on the one hand and the erosion of monetary sovereignty by private stablecoin initiatives on the other [2]. The cautionary tale of Facebook's Libra project, rebranded as Diem before its dissolution in 2022, illustrates the regulatory resistance to private multinational currency schemes [3], yet the underlying demand for digital payment alternatives persists. Payment giants Visa and Mastercard control over 70% of European card transactions [4], extracting costs from the payments infrastructure while concentrating transaction data in non-European jurisdictions [5]. In the near future, Big Tech platforms will possess (and sometimes they already do) both the technological capability and the user bases to launch proprietary payment systems that could further fragment European monetary sovereignty, a prospect that lends urgency to the digital euro initiative. The defense dimension of digital monetary infrastructure has received comparatively little attention in the academic liter-

ature, yet it is increasingly salient in European institutional discourse. A payment system dependent on non-European infrastructure represents a structural vulnerability in scenarios of geopolitical tension or economic coercion: the *weaponisation* of financial infrastructure [6] [7] [8] [9] demonstrates that payment systems are instruments of governance as much as instruments of commerce. The digital euro is thus positioned not merely as a technological innovation but as an instrument of strategic autonomy, designed to ensure that European citizens retain access to central bank money in an increasingly digital economy while preserving the euro's international role [10]. The success or failure of this ambition depends critically on the design choices that remain the subject of careful reflection and on actual adoption by citizens. The strategic autonomy dimension acquires particular importance in the context of the present analysis. A digital euro whose online privacy model fails to materially exceed the protections offered by non-European payment incumbents does not merely risk adoption failure; it risks reproducing the very dependency structure it was designed to displace. If Big Tech payment platforms offer privacy profiles functionally equivalent to those of the digital euro, the competitive rationale for a sovereign European payment infrastructure is substantially weakened. The graduated framework proposed is therefore not only a response to GDPR proportionality requirements but also a design choice with strategic implications: a digital euro offering demonstrably superior and constitutionally grounded privacy protections for routine transactions would create a value proposition unavailable to non-European providers bound by different regulatory regimes. This argument further extends into the domain of defensive digital sovereignty: a privacy-preserving CBDC architecture that generates no persistent linkable records for low-value transactions simultaneously reduces the intelligence value of any infrastructure compromise, thereby diminishing both the incentive for adversarial data collection and the damage potential of successful intrusion. Privacy-by-design in the transaction layer thus operates at once as a constitutional requirement, an adoption precondition, and a structural contribution to the resilience of European financial infrastructure against hybrid threats and cyberattacks.

The digital euro project has evolved through three distinct phases. The investigation phase, spanning October 2021 to October 2023, established foundational design principles and conducted extensive stakeholder consultations across 27 member states [10]. In this phase, which saw a high level of public participation, privacy emerged as one of the predominant concerns, cited by 59% of small merchants [11]. The preparation phase, launched in November 2023 and extended through October 2025, focused on rulebook development, prototyping and provider selection. The ECB Governing Council decision of 18 October 2025 authorised continuation beyond the initial two-year preparation timeline, reflecting both the technical complexity and the political sensitivities involved [10]. Critical milestones achieved during this phase include the selection of five external vendors who have signed framework agreements for the provision of front-end solutions (apps and

software development kits (SDKs)), offline functionality, alias lookup, risk and fraud management, and secure information exchange, as well as the designation of six national central banks for the development of the internal settlement and issuance components [12]. The rulebook development process exemplifies the governance complexity of the initiative: over 2,000 comments were processed from more than 50 participants representing over 30 organisations, including banks, payment service providers, merchants and consumer advocacy groups [10]. Implementation cost estimates reveal the scale of transformation required, with European banks facing projected investments between 4 billion and 5.8 billion euros to integrate digital euro infrastructure, while ECB development expenditures through 2028/2029 are projected at 1.3 billion euros, with annual operating costs stabilising at approximately 320 million euros thereafter [10] [12]. The legal foundation for the digital euro remains under construction. The European Commission’s proposed regulation on establishing the digital euro as legal tender was anticipated in Q4 2023 but has been delayed amid disagreements over privacy provisions, holding limits and the distribution model [10]. The proposed framework must reconcile multiple regulatory regimes: the Markets in Crypto-Assets Regulation establishing stablecoin oversight, the Fifth and Sixth Anti-Money Laundering Directives imposing customer due diligence requirements, and the General Data Protection Regulation setting privacy standards. Although a thorough examination of the implications falls outside the scope and methodology of the present work, it is also worth mentioning that the digital euro raises fundamental legal issues concerning its ontological status. Unlike physical cash, which exists as anonymous bearer instruments with no centralised ledger, digital currency inherently generates transaction records. Whether these records constitute personal data subject to GDPR protections or monetary system infrastructure exempt under public interest derogations, remains contested in the literature [3]. This ambiguity has profound implications: if digital euro transactions are deemed personal data, users gain rights to access, rectification, erasure and data portability that may conflict with Anti-Money Laundering retention requirements and monetary policy functions. The tension between GDPR principles and Financial Action Task Force recommendations exemplifies the regulatory collision at the heart of the initiative. GDPR mandates data minimisation and purpose limitation, requiring that only necessary personal data be collected for specified purposes, whereas Financial Action Task Force Recommendation 16 requires payment service providers to collect and retain complete originator and beneficiary information for transactions above 1,000 euros, with suspicious transaction reporting obligations extending to lower thresholds. A further dimension of this debate concerns whether access to central bank money might be recognised as a component of economic citizenship, and whether such recognition would entail a right to transactional privacy analogous to the privacy afforded by physical cash. The European Court of Justice has recognised that the right to privacy encompasses protection against comprehensive profiling and

behavioural tracking [13], though whether these precedents extend to payment system design remains an open question.

III. TECHNICAL ARCHITECTURE AND THE INTERNATIONAL CBDC LANDSCAPE

A critical clarification concerns the role of blockchain technology in the digital euro. Contrary to some public speculation, the ECB has explicitly stated that the digital euro will not be based on distributed ledger technology (DLT), though it incorporates key design principles inspired by DLT research [10]. This architectural choice which merits careful examination, reveals fundamental decisions about control, scalability and governance that distinguish the digital euro from cryptocurrency-based payment systems. The ECB’s rationale for not adopting a public blockchain foundation rests on several pillars. The first is scalability: public blockchains such as Bitcoin process approximately 7 transactions per second and Ethereum between 15 and 30 [14], whereas the digital euro must support potentially 340 million European citizens conducting retail transactions with sub-second settlement finality [11] [10]. The second concerns governance control, as public blockchains distribute decision-making authority across decentralised validator networks while the ECB requires centralised monetary policy control and the ability to implement legal requirements instantaneously. The third relates to privacy limitations, since public blockchain transparency conflicts with GDPR requirements for data minimisation and the right to erasure [15]. The fourth is timeline realism: developing a production-grade permissionless blockchain capable of Central Bank Digital Currencies-scale throughput would require a decade or more of research, exceeding realistic policy windows [16].

Instead, the ECB has designed a hybrid centralised architecture incorporating DLT principles [17]: geographical distribution across multiple regions with redundant servers, decentralised settlement execution [18] [19] allowing National Central Banks and payment service providers to operate nodes, atomicity ensuring transactions either complete fully or fail entirely, immutability of committed transactions providing audit trail integrity, and distributed consensus mechanisms among Eurosystem components for system state validation [10]. The resulting infrastructure might be characterised, borrowing from the taxonomy common in blockchain research, as functionally analogous to a private permissioned ledger: a single trusted authority, the Eurosystem, maintains the canonical state of the system, while selected participants such as NCBs and PSPs interact with it through controlled interfaces. The distinction from a true distributed ledger lies in the absence of a trustless consensus mechanism among independent validators; trust is instead vested institutionally in the ECB. This architectural choice is consequential for the privacy analysis that follows, because it means that the Eurosystem retains, by design, the technical capacity to access the entirety of the pseudonymised transaction ledger, a capacity that would not exist in the same form in a genuinely decentralised system. The technical infrastructure includes three regional implementations with multiple servers, a PSD2-compliant REST interface architecture, AI-based fraud detection systems based

on aggregated pseudonymized data, and the use of secure elements (eSE/eSIM) for secure management of digital keys on devices. [10]. This distinction is important: the digital euro is fundamentally different from a payment system built on blockchain or distributed ledger technology, and treating them as equivalent mischaracterises the architecture the ECB has chosen to pursue. The ECB's approach to privacy combines architectural guarantees with cryptographic techniques. The core principle is pseudonymization: the ECB processes transactions using pseudonymous identifiers that cannot be directly linked to individuals without access to additional information held separately by the PSPs. When a user initiates a transaction, the PSP validates the identity and authorization, but only transmits a pseudonymous identifier to the settlement system. The ECB ledger records the pseudonymized transaction data required for settlement and fraud detection, but not the actual identities of the parties. This design ensures that the Eurosystem cannot link transactions to identified individuals, providing an architectural guarantee of privacy. Offline functionality represents a parallel privacy mechanism designed to replicate "cash-like anonymity" [20] for proximate transactions. The ECB has developed specifications for secure element hardware embedded in cards or smartphones, enabling peer-to-peer digital euro transfers without internet connectivity or intermediary validation. These offline transactions rely on cryptographic protocols executed entirely within secure hardware, with transaction records inaccessible to both the Eurosystem and payment service providers until the device synchronises with the network. The offline model imposes important constraints, however: holding limits to mitigate loss and theft risks, periodic online synchronisation requirements to prevent double-spending, and merchant acceptance requiring compatible hardware [10]. These limitations suggest that offline capability will function primarily as a fallback rather than the primary mode of usage. Without purporting to undertake a comparative legal analysis, which would require a systematic examination of the domestic legal orders concerned, the following discussion draws upon selected international CBDC experiences insofar as they illuminate adoption dynamics and privacy design choices of relevance to the European context. The diversity of implementations globally reveals distinct philosophical and political orientations toward the privacy-control tradeoff, and it is instructive to situate the digital euro within this spectrum. China's digital yuan [1] represents what might be termed a surveillance-maximalist model: the People's Bank of China retains comprehensive access to all transaction data, enabling real-time monitoring of economic activity and individual financial behaviours [21]. While some privacy protections exist for small transactions in pilot programmes, the fundamental architecture grants the state comprehensive visibility into citizen financial activity. At the other end of the spectrum, the Swedish Riksbank e-krona research project, finalized in 2023 [22], examined designs where even the central bank would have limited access to transaction-level data, relying on statistical aggregates for monetary policy purposes, while preserving the privacy of individual transac-

tions by opting for the existence of an underlying ledger to record transactions and verify the rightful owner, concluding in favor of de facto traceability of e-krona transactions. [23]. The Bahamas Sand Dollar, operational since 2020 as the world's first fully implemented retail Central Bank Digital Currencies, represents another model focused on financial inclusion. Privacy protections in the Sand Dollar are contingent on anti-money laundering compliance, with all transactions linked to KYC-verified identities [24]. Adoption rates are extremely low [25], suggesting that implementing a Central Bank Digital Currencies without a compelling user value proposition could result in limited adoption, regardless of technical sophistication [24] [25]. Nigeria's eNaira, launched in October 2021, also reported similarly disappointing data, with adoption reaching around 5% but with extremely limited real-world wallet operation [26]. These cases, while operating in different economic and institutional contexts than those in Europe, provide empirical data that deserves careful consideration. The Brazilian instant payment system Pix provides valuable empirical evidence on adoption dynamics, although it's important to note that Pix is not a Digital Currencies, but rather a centralized instant payment infrastructure managed by the Central Bank of Brazil. Launched in November 2020, Pix has achieved extraordinary penetration: over 150 million registered users, equivalent to 70% of the adult population, in 24 months, over 3 billion transactions processed in 2023, and over 900 participating financial institutions. [27] Pix's success derived from deliberate design choices prioritising adoption. The system mandates participation by all deposit-taking institutions above minimum size thresholds, eliminating coordination problems that plague voluntary payment networks. Transaction fees are zero for individuals and negligible for merchants, creating strong incentives to shift from card-based payments. User experience innovations such as QR code scanning, key-based addressing using phone numbers rather than complex bank details, and instant irrevocable settlement eliminating chargeback risks proved decisive in driving uptake [28]. The social impact has been significant: approximately 45 million Brazilians who previously had no access to a bank account or had an insufficient one have gained access to digital payment features through Pix, effectively bringing cash-only users to the use of digital systems. Yet Pix operates under full traceability: every transaction is linked to identified parties, and the Central Bank of Brazil maintains complete transaction visibility for regulatory purposes. Privacy concerns have been raised but have not significantly impeded adoption, suggesting that in the Brazilian context, convenience and financial inclusion outweigh privacy preferences [28]. The critical question for the digital euro is whether lessons from Pix can be transposed to a European context where privacy preferences appear substantially stronger. German survey research indicates that European citizens place significantly higher value on financial privacy, potentially reflecting different historical experiences with surveillance and distinct cultural attitudes toward state authority [29]. This divergence could suggest that a fully traceable model, however successful in Brazil, might

face resistance in Europe. At the same time, Pix demonstrates several principles that could be applicable: mandatory participation by major financial institutions, zero or minimal fees, simplified user experience, instant settlement and government commitment signalling permanence. The academic literature on central bank digital currency design has developed rapidly, but debate persists on the macroeconomic effects of the introduction of central bank digital currencies, on banking disintermediation and monetary policy transmission [30], on cryptographic privacy techniques, including zero-knowledge proofs and blind signatures [31], on comparative institutional analysis of central bank digital currency designs in different jurisdictions [16], on legal and regulatory frameworks [32], and on user acceptance research.

IV. PRIVACY AS A DETERMINATIVE DESIGN VARIABLE

The preceding overview of the digital euro’s institutional trajectory and its international counterparts reveals a recurring structural tension that may prove determinative for the project’s viability in the European context. This tension concerns the relationship between transactional privacy and anti money laundering compliance, a relationship that the current architectural design addresses through uniform pseudonymisation but that, as will be shown, may require a more nuanced and differentiated approach if the digital euro is to achieve meaningful adoption among European citizens whose privacy expectations differ materially from those observed in other jurisdictions. The present section examines this tension by identifying an architectural asymmetry in the current ECB design and by reviewing the empirical evidence on European privacy preferences that renders this asymmetry problematic. A particularly interesting asymmetry emerges from the current architectural design of the digital Euro when considering the differential treatment afforded to offline and online transactions. Offline payments are designed to provide a cash-like level of privacy, with transaction data remaining entirely within the secure element of the user’s device and inaccessible to both the Eurosystem and payment service providers. Online transactions, by contrast, receive only the protection of pseudonymisation, a mechanism that, while preventing the ECB from directly identifying individuals, still generates a persistent and structured data trail at the level of the payment service provider. Nevertheless it is precisely in the online environment that the overwhelming majority of economically and socially significant transactions take place today, from the payment of utilities and insurance premiums to the purchase of medical services, cultural goods and subscription platforms. These transactions collectively constitute what one might describe as the digital imprint of an individual’s financial and personal life, and it is this domain that the current model leaves most exposed. One might reasonably ask, therefore, whether a privacy architecture that reserves its strongest protections for the least-used modality whilst affording only structural pseudonymisation to the channel through which citizens conduct the vast majority of their economically meaningful activ-

ity can genuinely be said to reflect the privacy expectations that European survey research has consistently documented.

Similar to the European study, the Deutsche Bundesbank study, which interviewed 2,012 German residents, reached similar conclusions regarding privacy. It found that 73% of German citizens “are particularly concerned about protecting their privacy. More than three-quarters of respondents rated the aspect of privacy in connection with the use of the digital euro as very important or important. Of those surveyed, 59% stated that the planned offline version of the digital euro, which is intended to provide a level of privacy protection similar to that of cash, was very important or important.” [33] This finding is noteworthy: it suggests that European citizens might adopt the digital euro if and only if it offers privacy protections perceived as substantially superior to current commercial payment methods. The issue of privacy, even in the context of digital transactions (both in person and online), does not appear to be a variable to be considered or balanced but, on the contrary, could represent the difference between the success and failure of the technology. A critical gap in the literature concerns the lack of concrete proposals for graduated privacy architectures that implement functional anonymity proportional to transaction risk and that are also applicable to remote transactions. Privacy is sometimes framed as a binary choice between complete anonymity and full traceability, without seriously exploring the intermediate positions [34] [35].

From a theoretical standpoint, the concept of contextual integrity, developed by Nissenbaum, offers a useful frame: information flows are appropriate when they match the norms of the context in which the information originates [36]. Routine retail transactions generally occur within a social context whose norms do not ordinarily demand identification; graduated anonymity architectures can be understood as an attempt to encode these contextual norms into payment system design, rather than imposing a uniform logic regardless of transaction context.

V. THE PRIVACY-COMPLIANCE TENSION: PROBLEM DEFINITION AND OPEN QUESTIONS

The digital euro confronts a tension between two legitimate but conflicting imperatives: preserving individual privacy rights comparable to physical cash, and ensuring regulatory compliance with anti-money laundering and counter-terrorism financing obligations. This tension is not merely technical but reflects deeper disagreements about the proper balance between individual liberty and collective security [37] [38]. The traceable-by-design architecture inherent in digital currency systems creates comprehensive transaction records without analogue in cash economies. Every digital euro payment generates data: the payer’s pseudonymous identifier, the payee’s identifier, the transaction amount, the timestamp, the merchant category and potentially the geographic location. Even when pseudonymised at the ECB level, payment service providers retain the ability to link these patterns to identified individuals, enabling granular profiling of consumer behaviour.

The risk of behavioural profiling and *chilling effects* merits serious consideration. If citizens know that every purchase, donation and financial interaction generates a permanent record accessible to payment providers and potentially to government authorities, rational actors may alter their behaviour to avoid surveillance [39]. This *chilling effect* extends beyond illegal activity to encompass entirely lawful but potentially stigmatised behaviours: purchasing political literature, supporting controversial social or political causes, accessing reproductive health services. The ECB’s current architecture, while it aims to prevent the central bank from conducting direct surveillance, does not prevent by design interested subjects from constructing detailed user profiles.

The centralisation of payment data creates further risks. A single data breach at a major payment service provider could expose the financial behaviours of millions of Europeans. A future government with authoritarian inclinations could mandate access to pseudonymised transaction databases, enabling retrospective surveillance. While current European governance structures provide protections (e.g. The GDPR regulation), institutional design arguably should account for potential future misuse [39].

The ECB and broader central banking community have explored various privacy-preserving technologies, though implementation remains largely experimental. Zero-knowledge proofs represent perhaps the most promising cryptographic innovation for privacy-preserving compliance. These protocols allow one party to prove possession of information satisfying certain criteria without revealing the information itself. Technically, this is achieved through cryptographic constructions, most notably zk-SNARKs (Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge), which generate compact, verifiable proofs without requiring interaction between the prover and the verifier. The Bank for International Settlements Innovation Hub has conducted experiments exploring the possibility of settling central bank digital currencies on distributed ledgers through Project Helvetia, demonstrating end-to-end integration with core banking systems and legal feasibility on privately operated DLT platforms [40], although these experiments have not addressed the distinct challenges of retail Central Bank Digital Currencies where sub-second latency requirements impose additional constraints. Beyond zero-knowledge proofs, threshold cryptography offers another approach through distributed trust, splitting cryptographic keys across multiple parties such that a predetermined number must cooperate to decrypt transaction data. This prevents any single entity, including the central bank, from unilaterally accessing user data [41]. Blind signature schemes, pioneered by Chaum in the 1980s, enable a payment authority to sign a digital token without seeing its contents, providing transaction-level unlinkability while preventing double-spending [42]. Hardware-based privacy using secure elements and trusted execution environments provides an architectural complement to cryptographic techniques, executing privacy-critical operations within tamper-resistant hardware isolated from the operating system [10]. An objective assessment of the digital euro’s cur-

rent trajectory suggests that adoption challenges will likely be substantial in the absence of design revisions, not because of design flaws but due to factors that could impact the end user. First, there is a real risk of inferior user experience compared to traditional private payment systems. Apple Pay, Google Pay, and contactless credit cards offer one-touch payments and instant confirmation, requiring no user training beyond touching a device. The digital euro may require installing a wallet app, identity verification, initial funding from bank accounts, and potentially different interaction models for online and offline contexts [10]. Unless the digital euro offers parity or superiority in terms of user experience, ordinary consumers will continue to use familiar methods [11].

The second factor concerns privacy as a potential differentiation axis. If the digital euro’s privacy model merely matches that of Visa or Mastercard, offering pseudonymisation from the ECB’s perspective but full visibility to payment service providers, it fails to create a compelling value proposition. Survey research consistently identifies privacy as a top-tier concern, yet the phenomenon labelled the “privacy paradox” complicates predictions: consumers desire privacy but often fail to act on these preferences when faced with convenience tradeoffs [43]. This does not render privacy irrelevant to adoption, but it suggests that privacy must be delivered as a default feature requiring no user effort or expertise.

The third factor is the network effects problem. Payment systems exhibit strong same-side and cross-side network effects, and Visa and Mastercard have global acceptance networks built over decades. The ECB’s proposed zero interchange fees could incentivise merchant adoption [11] [10], but this advantage may be offset by integration costs and uncertain transaction volumes. Empirical evidence from operational Central Bank Digital Currencies reinforces these concerns: both the Bahamian Sand Dollar and Nigeria’s eNaira received only marginal adoption despite meeting the threshold conditions that might ordinarily be expected to drive uptake, namely legal tender status and national availability [26]. These outcomes, though occurring in contexts that differ considerably from Europe’s, provide grounds for caution regarding assumptions about automatic Central Bank Digital Currencies adoption.

VI. TOWARD GRADUATED FUNCTIONAL ANONYMITY: AN EXPLORATORY FRAMEWORK

A. Applying Functional Anonymity to Central Bank Digital Currencies

The preceding analysis could suggest that the binary framing of the privacy debate, complete anonymity versus full traceability, may be insufficiently nuanced. One possible conceptual alternative is what could be termed *functional anonymity* [44]: a privacy framework that aims to replicate the practical privacy characteristics of physical cash in digital systems without demanding identical technical properties. The underlying observation is that physical cash itself exhibits graduated privacy in practice. A consumer purchasing a product with a small sum using cash has complete anonymity,

yet large cash payments trigger reporting requirements in many jurisdictions, merchants may request identification, and physically transporting large amounts attracts attention and risk. This graduated practical privacy has evolved organically as a *de facto* balance between individual autonomy for routine transactions and regulatory oversight of high-risk activity. In Italy, for example, every payment above €4,999.99 must be made using a traceable payment system. The idea that privacy protections in digital currency systems can be calibrated proportionally to the value and risk of the transaction is not entirely unprecedented in the literature, although concrete architectural proposals remain scarce. The IMF, in a 2024 Fin-Tech Note, discusses multi-layered due diligence approaches for CBDCs, while examples such as China (e-CNY) and Nigeria (eNaira) implement tiered KYC requirements that enable managed anonymity for low-value transactions. The Bank for International Settlements’ Innovation Hub, through Project Tourbillon, demonstrated the technical feasibility of blind signature schemes that enable payer anonymity for retail CBDC transactions, while the ECB’s own EUROchain proof of concept, developed with eighteen national central banks, tested a voucher-based anonymity model in which AML thresholds determined the ceiling of unlinkable transactions rather than selectively intervening on individual payments. More recently, academic literature has begun to formalize dual-access CBDC models that combine account-based and token-based modes, offering differentiated levels of anonymity that users can select based on their preferences. The European legal basis for this approach stems from Article 52, paragraph 1, of the Charter of Fundamental Rights of the European Union, which establishes that any limitation on the exercise of the rights recognized by the Charter must respect the principle of proportionality and may be imposed only to the extent necessary and genuinely meets objectives of general interest recognized by the Union. This constitutional requirement finds concrete operational expression in the principle of data minimization enshrined in Article 5, paragraph 1, letter c), of the GDPR, which requires that the processing of personal data be adequate, relevant, and limited to what is necessary in relation to the purposes for which they are processed. Applied to payment system architecture, this principle suggests that anti-money laundering oversight measures that generate permanent records of transactions should be proportionate to the actual money laundering risk posed by each category of transaction, rather than being uniformly imposed across all value thresholds [45] [46] [47] [48] [49]. The concept of functional anonymity proposes deliberately engineering this gradualism into digital currency architecture rather than treating all digital euro transactions as possessing identical privacy characteristics. The philosophical foundation rests on the proportionality principle, well established in European constitutional law, according to which privacy intrusions by state authorities must be proportionate to legitimate objectives, imposing minimal necessary restrictions on fundamental rights. Applied to payment systems, proportionality would suggest that anti money laundering compliance measures should scale with money laundering risk, which correlates with transaction value and pattern characteristics. It is worth noting that functional

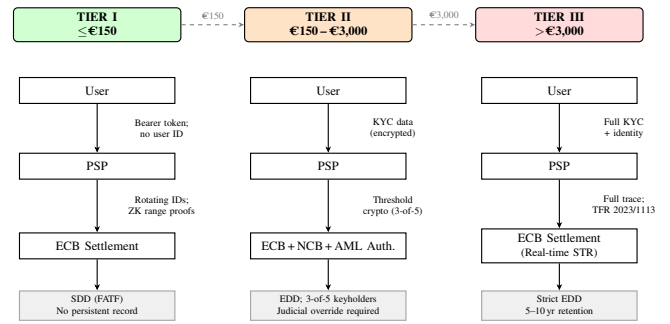


Fig. 1: Three-tier graduated functional anonymity architecture for the digital euro. Tier-switching is determined by transaction value, evaluated at the PSP layer. ZK: zero-knowledge proofs; SDD: simplified due diligence; EDD: enhanced due diligence; TFR: Transfer of Funds Regulation (EU) 2023/1113; STR: suspicious transaction report.

anonymity differs from mere pseudonymisation in a critical respect: pseudonymisation protects identity from unauthorised parties but preserves linkability by authorised parties, whereas functional anonymity would seek to make certain transactions practically unlinkable even to authorised observers, subject to specific legal override mechanisms. The distinction matters because pseudonymisation enables comprehensive profiling by payment service providers and retrospective surveillance by authorities with access to mapping keys, whereas functional anonymity at appropriate levels would aim to prevent such profiling entirely for lower-risk transactions, creating a non-revocable privacy implementation by design that maintains appropriate levels of confidentiality while ensuring public safety.

B. A Possible Three-Tier Architecture

It is important to situate this proposal against the current state of the ECB’s own design choices. The architecture described in the preparation phase closing report of October 2025 offers what amounts to a binary privacy model: offline transactions benefit from strong anonymity guarantees, with payment data remaining on the device and inaccessible to both PSPs and the Eurosystem, while all online transactions, regardless of their value or nature, are subject to uniform pseudonymisation with full transaction visibility at the PSP level. No mechanism is envisaged within the current design to differentiate the privacy treatment of a two-euro purchase from that of a three-thousand-euro transfer in the online channel. This absence of proportionality in the online privacy model has been noted in recent critical assessments of the digital euro proposal, with Cannataci et al. observing that the online component essentially replicates the data exposure profile of existing commercial payment systems without delivering meaningful additional privacy protections. The three-tier framework outlined below is therefore proposed as an alternative that the current ECB design does not contemplate, aiming to extend the principle of proportionality, already operative in European constitutional law and data protection

regulation, into the transaction-level architecture of the online digital euro [50]. Without claiming to offer a definitive or prescriptive design, one possible approach to operationalising graduated functional anonymity might involve a three-tier structure calibrated to transaction value and Anti-Money Laundering risk. The thresholds suggested here are illustrative and would require empirical validation on large scale and political deliberation, but they draw upon existing Anti-Money Laundering regulatory boundaries, Eurozone cash transaction patterns and the ECB’s own considerations regarding holding limits. The first tier (Table I), which could cover transactions up to approximately 150 euros, would aim to provide enhanced privacy approaching that of physical cash. The design principle would be that everyday payments for daily necessities should generate minimal durable records linking specific individuals to specific transactions. Technical implementation could draw upon offline-first architecture, bearer digital instruments using cryptographic tokens signed by the ECB but containing no user-identifying information, and rapidly rotating pseudonymous identifiers for online transactions achieving unlinkability across individual payments. Zero-knowledge range proofs could enable users to demonstrate that transaction amounts fall within the tier without revealing exact values. The threshold of 50-150 euros represents a possible midpoint between the ECB’s current consideration of micropayment anonymity and the Financial Action Task Force thresholds for simplified due diligence: Eurostat data indicates that approximately 85% of cash transactions in the Eurozone fall below 50 euros, and might be reasonably possible to extend the privacy zone to capture a broader range of routine retail activity [10]. The second tier (Table I), covering transactions between approximately 150 and 3,000 euros, could implement pseudonymity with additional privacy-preserving safeguards. It would offer complete Know Your Customer (KYC) functionalities, meeting Anti-Money Laundering Directive requirements, but the linkage between identity and transaction history would be protected by cryptographic access controls requiring multi-party authorisation. Threshold cryptography distributing decryption capability across multiple independent authorities, for instance requiring three of five keyholders including the ECB, the relevant National Central Bank and the national Anti-Money Laundering authority, could prevent any single entity from unilaterally accessing transaction histories. This design would operationalise the principle that privacy is not absolute but subject to legal override with procedural safeguards: law enforcement seeking to investigate suspicious transactions would need judicial authorisation and coordination with multiple independent authorities, creating audit trails and institutional checks. Homomorphic encryption techniques could enable fraud detection algorithms to operate on encrypted data, identifying suspicious patterns without routinely decrypting individual transactions. The third tier (Table I), for transactions exceeding 3,000 euros, would operate under full Anti-Money Laundering compliance with the constitutional safeguards already provided by GDPR. This threshold aligns with EU Anti-Money Laundering Directive requirements for enhanced

customer due diligence and corresponds to cash payment restrictions in multiple member states [10]. All transactions would be fully traceable with real-time suspicious transaction monitoring. The EU Transfer of Funds Regulation (EU) 2023/1113 would apply to fund transfers above the relevant thresholds, requiring payment service providers to transmit complete originator and beneficiary information accompanying all fund transfers, a requirement that the Financial Action Task Force Travel Rule had already established at the international level for transactions above 1,000 USD. Full traceability at this level would not entail absence of privacy protections: GDPR provisions for data minimisation, purpose limitation and restrictions on automated decision-making would continue to apply, and transaction data would be processable only for specified legitimate purposes. Whereas the second tier preserves identity linkage as a cryptographically protected exception requiring multi-party authorisation to unlock, the third tier inverts this logic entirely: full traceability becomes the operative default, with privacy protections functioning not as a structural barrier to identification but as a set of downstream constraints governing the permissible purposes and conditions for the use of data that is, by design, already accessible. Unlike the ECB’s current binary model, which reserves strong privacy protections for the offline channel irrespective of transaction value, the proposed framework relocates the discriminating variable from the payment medium to the transaction amount, extending meaningful privacy protections to the online channel where the overwhelming majority of economically significant transactions occur. It should be emphasised that this three-tier structure is presented as one possible configuration among several that might be explored. The specific thresholds, the cryptographic implementation choices, and the governance arrangements would all require extensive deliberation, technical testing and political negotiation. The point is not to prescribe a solution but to suggest that a graduated approach warrants serious consideration as an alternative to the current uniform pseudonymisation model.

TABLE I: Proposed three-tier graduated functional anonymity framework for digital euro privacy architecture. Thresholds are illustrative. ZK: zero-knowledge; SDD: simplified due diligence; FATF: Financial Action Task Force; KYC: know your customer; EDD: enhanced due diligence; STR: suspicious transaction report; TFR: Transfer of Funds Regulation (EU) 2023/1113; PQC: post-quantum cryptography.

Tier	Range	Privacy level	Mechanism	AML
I	$\leq \text{€}150$	Cash-like anonymity; unlinkable, no persistent record	Bearer tokens; rotating IDs; offline-first; ZK range proofs	SDD (FATF)
II	$\text{€}150\text{--}\text{€}3\text{k}$	Protected pseudonymity; identity linkage via multi-party auth + judicial override	Full KYC; threshold crypto (3-of-5); homomorphic fraud detection; blind sigs	EDD; conditional court access
III	$>\text{€}3\text{k}$	Full traceability; GDPR as downstream constraint on data use	Real-time STR; TFR 2023/1113; FATF Travel Rule; NIST PQC	Strict EDD; 5–10 yr retention

Even if a graded functional anonymity model were deemed desirable in principle, significant implementation challenges would currently remain. The first set of challenges is technical. The cryptographic protocols underlying layered anonymity, particularly zero-knowledge proofs and threshold cryptography, impose a computational overhead that could prove difficult to reconcile with sub-second payment processing on a European scale. While acceptable for individual transactions, these latencies could create bottlenecks during peak usage. The second set of challenges concerns regulatory coordination. The EU’s legal architecture distributes monetary policy authority to the ECB while retaining national competence over Anti-Money Laundering enforcement, criminal procedure and data protection implementation. If threshold decryption were to require judicial authorisation, questions would arise about which national court holds jurisdiction for cross-border transactions, how conflicting privacy standards across member states would be reconciled, and how regulatory arbitrage might be prevented. The Anti-Money Laundering regulatory landscape remains fragmented despite harmonisation efforts, and the absence of fiscal union complicates matters further. The third challenge, perhaps the most significant, is the risk of privacy erosion through the gradual and unplanned expansion of a technology’s scope or reach. Historical experience with modern technologies demonstrates a consistent pattern: systems designed with limited scope and robust security measures gradually expand to broader applications. Digital euro transaction data as currently conceived would represent sensitive and exposed information that future attackers could exploit. The issue of data retention, however, presents an ongoing tension, as anti-money laundering regulations require transaction logs to be retained for 5–10 years. A fourth challenge concerns future-proofing the technology. One vulnerability that is only hinted at today, for example, concerns quantum computing. The security of proposed cryptographic protocols depends on mathematically complex assumptions that quantum computers running Shor’s algorithm could ultimately break [51]. The “collect now, decrypt later” threat incentivizes adversaries to record encrypted transaction data now and then decrypt it once quantum computers become available. Post-quantum algorithms standardized by NIST offer potential solutions, but they introduce larger key sizes and higher computational costs, making the transition of the entire digital euro ecosystem extremely complex. Finally, the interoperability of digital currencies between central banks raises questions for which there are still no clear answers. If the digital euro were to implement gradual anonymity while other jurisdictions maintain full traceability, the processing of cross-border payments would need to be resolved, as two potentially incompatible protocols would be interfacing. International cooperation frameworks for the interoperability of digital currencies between central banks are still in their infancy, and the lack of agreed-upon standards for privacy levels, anti-money laundering compliance, and data sharing creates the risk of fragmentation.

VII. CONCLUSION

This paper has examined the European Central Bank’s digital euro project through the lens of the privacy-compliance

TABLE II: Comparison Between the ECB Current Privacy Model and the Proposed Graduated Functional Anonymity Framework

Dimension	ECB Current Model	Proposed Graduated Framework
Discriminating variable	Payment medium: offline = strong anonymity; online = uniform pseudonymisation regardless of value.	Transaction value and AML risk: privacy scales proportionally with money laundering risk across three tiers.
Online privacy level	Uniform pseudonymisation for all online transactions: €2 and €2,900 receive identical treatment.	Tier I: cash-like anonymity. Tier II: protected pseudonymity with cryptographic access controls. Tier III: full traceability.
Offline privacy level	Strong anonymity: data stays in secure element; inaccessible to PSPs and Eurosystem.	Preserved at Tier I equivalent; no persistent linkable records.
AML mechanism	KYC at PSP onboarding; pseudonymous IDs to ECB; PSP retains full identity-mapping.	Tier I: SDD (FATF). Tier II: KYC + threshold crypto (3-of-5) + judicial override. Tier III: real-time STR + TFR 2023/1113.
Privacy instrument	Pseudonymisation: unlinkable to ECB but linkable by PSPs.	Tier I: bearer tokens + rotating IDs. Tier II: threshold crypto; homomorphic fraud detection. Tier III: GDPR as downstream constraint.
Constitutional basis	Art. 5(1)(c) GDPR applied uniformly; no risk differentiation.	Art. 52(1) CFR (proportionality) applied by risk tier; downstream expression in Art. 5(1)(c) GDPR.
Identity access model	PSP holds full pseudonym-to-identity mapping; ECB cannot link without PSP cooperation.	Tier I: no mapping created. Tier II: multi-party authorisation + judicial oversight. Tier III: accessible under AML/legal process.

tension, drawing upon institutional documentation, comparative analysis of international Central Bank Digital Currencies implementations and empirical evidence on European privacy preferences. The analysis reveals that the current uniform pseudonymisation model while technically operable, leaves unresolved a series of questions regarding user adoption incentives, privacy differentiation from commercial payment systems, and long-term resilience against surveillance expansion. Beyond its constitutional and regulatory dimensions, the proposed framework carries implications for European digital sovereignty and defensive resilience: a digital euro whose privacy architecture minimises persistent linkable records for routine transactions simultaneously limits the data available for exploitation by hostile actors. The concept of graded functional anonymity has been introduced as a possible alternative framework, calibrating privacy protections proportionally to

transaction value and money laundering risk through a three-tier architecture (Table I). The proposal is not technically focused and is intended to contribute to the ongoing debate rather than prescribe specific design choices. Available empirical evidence, ranging from the disappointing adoption figures for the Bahamian Sand Dollar and the Nigerian eNaira to calls for conditional privacy expressed in European surveys, suggests that the question of how to structure privacy protections in the digital euro is not merely a technical detail, but a factor that could prove decisive for adoption outcomes. Significant challenges remain, including technical scalability, regulatory coordination among heterogeneous Member States, the risks of privacy erosion, the threats of quantum computing, and interoperability between central bank digital currencies. These challenges are not invoked to suggest that graded anonymity is unattainable, but rather to indicate areas where further research, experimentation, and institutional deliberation are most urgently needed. Future research could productively focus on the empirical validation of graded privacy models through pilot programs, the legal analysis of how GDPR data protection requirements interact with the Financial Action Task Force's (FATF) anti-money laundering recommendations in digital currency contexts, and the development of post-quantum cryptographic protocols suitable for retail payment environments. The digital euro represents a potentially transformative initiative for the European monetary infrastructure. How its privacy architecture and implementation are designed will influence not only the dynamics of adoption in Europe but also the broader international trajectory of central bank digital currencies.

REFERENCES

- [1] E. J. Fullerton and P. J. Morgan, "The people's republic of china's digital yuan: Its environment, design, and implications," ADBI Working Paper, Tech. Rep., 2022.
- [2] A. Argentiero, "Criptovalute e sovranità monetaria: tra innovazione, rischio e regolamentazione," *De Iustitia et Iure-Legal journal of foundations and discussions-Open Access*, vol. 1, no. 2, 2025.
- [3] D. A. Zetzsche, D. W. Arner, and R. P. Buckley, "Decentralized finance," *Journal of Financial Regulation*, vol. 6, no. 2, pp. 172–203, 2020.
- [4] D. Bruggink and G. Lepecq, "The hidden costs of card payments," *CashEssentials*, December, 2016.
- [5] E. van Praag, "European payments: Faster, cheaper, more digital and more european, but leave no one behind," *Cheaper, More Digital and More European, but Leave No One Behind* (August 09, 2024), 2024.
- [6] J. D. Caytas, "Weaponizing finance: Us and european options, tools, and policies," *Colum. J. Eur. L.*, vol. 23, p. 441, 2016.
- [7] A. Nölke, "The weaponization of global payment infrastructures: A strategic dilemma," SAFE White Paper, Tech. Rep., 2022.
- [8] C. Westermeier, "The digital euro: A materialization of (in) security," *Review of international political economy*, vol. 31, no. 5, pp. 1569–1592, 2024.
- [9] M. F. Sharaf, A. M. Shahan, E.-H. I. Mansour, and N. N. Elboghady, "Financial resilience in a fragmented system: assessing russia's spfs and the sustainability of global payment networks," *International Economics and Economic Policy*, vol. 23, no. 2, p. 36, 2026.
- [10] European Central Bank, "Progress on the preparation phase of a digital euro: Closing progress report," European Central Bank, Frankfurt am Main, Germany, Tech. Rep. QB-01-25-249-EN-N, 10 2025. [Online]. Available: <https://www.ecb.europa.eu>
- [11] Ipsos European Public Affairs, "Ecb digital euro user research," European Central Bank, Frankfurt am Main, Germany, Research Report, 2025, research commissioned by the European Central Bank. [Online]. Available: <https://www.ecb.europa.eu>
- [12] European Central Bank, "Digital euro," https://www.ecb.europa.eu/euro/digital_euro/html/index.en.html, 2026, accessed: February 27, 2026.
- [13] N. Eder, "Privacy, non-discrimination and equal treatment: developing a fundamental rights response to behavioural profiling," in *Algorithmic Governance and Governance of Algorithms: Legal and Ethical Challenges*. Springer, 2020, pp. 23–48.
- [14] S. Rouhani and R. Deters, "Performance analysis of ethereum transactions in private blockchain," in *2017 8th IEEE international conference on software engineering and service science (ICSESS)*. IEEE, 2017, pp. 70–74.
- [15] U. Pagallo, E. Bassi, M. Crepaldi, and M. Durante, "Chronicle of a clash foretold: Blockchains and the gdpr's right to erasure," in *Legal Knowledge and Information Systems*. Ios Press, 2018, pp. 81–90.
- [16] R. Auer, G. Cornelli, and J. Frost, "Rise of the central bank digital currencies: Drivers, approaches and technologies," Bank for International Settlements, BIS Working Papers 880, August 2020.
- [17] M. Palmirani, C. Bomprezzi, and L. Papalia, "The use of blockchain for legislative simplification and tracking," in *2025 Eleventh International Conference on eDemocracy & eGovernment (ICEDEG)*. IEEE, 2025, pp. 97–105.
- [18] A. Arruzzoli, M. Zichichi, M. Palmirani, L. Papalia, C. Bomprezzi, and S. Ferretti, "A multi-chain approach to transparent and accountable legislative processes," 2026.
- [19] M. Palmirani, L. Papalia, C. Bomprezzi, A. Arruzzoli, M. Zichichi, S. Ferretti *et al.*, "Multi-level architecture for separation of powers in legislative process," in *Proceedings of the Seventh Distributed Ledger Technology Workshop (DLT2025)*, 2025, pp. 1–13.
- [20] F. Panetta, "The cost of not issuing a digital euro," Speech at the CEPR-ECB Conference: The macroeconomic implications of central bank digital currencies, Frankfurt am Main, November 2023, banca d'Italia. Available at: https://www.ecb.europa.eu/press/conferences/shared/pdf/20231123_cepr_ecb/Panetta_speech.en.pdf.
- [21] M. Chorzempa, "Promise and peril of digital money in china," *Cato J.*, vol. 41, p. 295, 2021.
- [22] Sveriges Riksbank, "E-krona," <https://www.riksbank.se/en-gb/payments--cash/e-krona/>, 2024, accessed: 2024-05-22.
- [23] —, "Monetary policy," <https://www.riksbank.se/en-gb/monetary-policy/>, 2024, accessed: 2024-05-22.
- [24] Alliance for Financial Inclusion. (2024, Jan.) Central bank digital currency: lessons from the bahamas. AFI Global. Opinion piece. [Online]. Available: <https://afi-global.org/opinion/central-bank-digital-currency-lessons-from-the-bahamas/>
- [25] K. Wenker, "Retail central bank digital currencies (cbdc), disintermediation and financial privacy: The case of the bahamian sand dollar," *FinTech*, vol. 1, no. 4, pp. 345–361, 2022.
- [26] B. Adedipe, "Adoption of the eNaira: issues and the way forward," *CBN Economic and Financial Review*, vol. 60, no. 4, pp. 171–180, Dec. 2022, article 13. [Online]. Available: <https://dc.cbn.gov.ng/efr/vol60/iss4/13>
- [27] A. Duarte, J. Frost, L. Gambacorta, P. Koo Wilkens, and H. S. Shin, "Central banks, the monetary system and public payment infrastructures: lessons from brazil's pix," Available at SSRN 4064528, 2022.
- [28] M. G. Schapiro, P. S. B. Mouallem, and E. G. Dantas, "Pix: explaining a state-owned fintech," *Brazilian Journal of Political Economy*, vol. 43, no. 4, pp. 874–892, 2023.
- [29] D. Potoglou, N. Robinson, S. Patil, B. Patrui, S. T. Trautmann, S. Arshad, and M. de Vries, "Privacy, security and surveillance: New insights into preferences of european citizens," *SSRN Electronic Journal*, Jan. 2014. [Online]. Available: <https://doi.org/10.2139/ssrn.2418346>
- [30] U. Bindseil, "Tiered CBDC and the financial system," European Central Bank, ECB Working Paper 2351, January 2020.
- [31] D. Chaum, C. Grothoff, and T. Moser, "How to issue a central bank digital currency," Swiss National Bank, SNB Working Papers 2021-03, 2021.
- [32] W. Bossu, M. Itatani, C. Margulis, A. Rossi, H. Weenink, and A. Yoshinaga, "Legal aspects of central bank digital currency: Central bank and monetary law considerations," International Monetary Fund, IMF Working Paper 20/254, November 2020.
- [33] Deutsche Bundesbank. (2024, Jun.) Bundesbank survey: Widespread acceptance of digital euro among general public. Deutsche Bundesbank. Press Release. [Online]. Available: <https://www.bundesbank.de/en/press/press-releases/bundesbank-survey-widespread-acceptance-of-digital-euro-among-general-public--933322>

- [34] A. Bernes *et al.*, “Privacy enhancing technologies, trasparenza e tutela della persona nell’ambiente digitale,” *MATERIALI E DOCUMENTI*, pp. 23–51, 2022.
- [35] L. Marchesin, “Oltre il consenso. i limiti dell’individual control model e prospettive per un approccio collettivistico alla sorveglianza contemporanea,” *Rivista italiana di informatica e diritto*, vol. 7, no. 1, pp. 361–383, 2025.
- [36] H. Nissenbaum, *Privacy in Context: Technology, Policy, and the Integrity of Social Life*. Stanford, CA: Stanford University Press, 2009.
- [37] R. Di Maria, “Il bilanciamento tra libertà civili e tutela della sicurezza interna alla luce della rule of law: la crisi del potere legislativo di fronte a quello giudiziario,” *University of Malta. Faculty of Laws*, 2008.
- [38] C. Di Francesco Maesa *et al.*, “Sicurezza vs libertà nello spazio di libertà, sicurezza e giustizia. un nuovo bilanciamento di valori operato dalla corte?” *Osservatorio della Giurisprudenza*, 2020.
- [39] M. Büchi, N. Festic, and M. Latzer, “The chilling effects of digital dataveillance: A theoretical model and an empirical research agenda,” *Big Data & Society*, vol. 9, no. 1, p. 20539517211065368, 2022.
- [40] Bank for International Settlements, Swiss National Bank, and SIX, “Project helvetia phase ii: Settling tokenised assets in wholesale cbdc,” Bank for International Settlements, Basel, Switzerland, BIS Innovation Hub Report, January 2022. [Online]. Available: <https://www.bis.org/publ/othp45.htm>
- [41] T. P. Pedersen, “A threshold cryptosystem without a trusted party,” in *Advances in Cryptology — EUROCRYPT ’91*, D. W. Davies, Ed. Berlin, Heidelberg: Springer Berlin Heidelberg, 1991, pp. 522–526.
- [42] D. Chaum, “Blind signatures for untraceable payments,” in *Advances in Cryptology – CRYPTO ’82*. Springer, 1983, pp. 199–203.
- [43] S. Barth and M. D. T. de Jong, “The privacy paradox: Investigating discrepancies between expressed privacy concerns and actual online behavior,” *Telematics and Informatics*, vol. 34, no. 7, pp. 1038–1058, 2017.
- [44] M. Elliot, K. O’Hara, C. Raab, C. M. O’Keefe, E. Mackey, C. Dibben, H. Gowans, K. Purdam, and K. McCullagh, “Functional anonymisation: Personal data and the data environment,” *Computer Law & Security Review*, vol. 34, no. 2, pp. 204–221, 2018, peer reviewed. Also available at: <https://kieronohara.com/article/functional-anonymisation-personal-data-and-the-data-environment/>. [Online]. Available: <https://doi.org/10.1016/j.clsr.2018.02.001>
- [45] N. Zhang, V. Budau, F. Solomon, K. Kao, M. Vucinic, and K. Miggiani, “Central bank digital currency data use and privacy protection,” 2024.
- [46] B. I. Hub, “Project tourbillon: exploring privacy, security and scalability for cbdc,” *Bank for International Settlements*, 2023.
- [47] E. Bank, “Exploring anonymity in central bank digital currencies,” *Focus*, December (4), vol. 1, no. 11, 2019.
- [48] R. A. Auer, R. Böhme, J. Clark, and D. Demirag, *Privacy-enhancing technologies for digital payments: mapping the landscape*. Bank for International Settlements, Monetary and Economic Department, 2025.
- [49] P. Sanz Bayón, “Current and future central bank digital currency (cbdc) projects,” in *Governance and Control of Data and Digital Economy in the European Single Market: Legal Framework for New Digital Assets, Identities and Data Spaces*. Springer, 2025, pp. 309–347.
- [50] J. Cannataci, B. Fehrensens, M. Gütschow, Ö. Kesim, and B. Lucke, “Digital euro: Frequently asked questions revisited,” *arXiv preprint arXiv:2601.18644*, 2026.
- [51] P. W. Shor, “Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer,” *SIAM Journal on Computing*, vol. 26, no. 5, pp. 1484–1509, 1997.